Summary in Graph

Exam Summary (GO Classes GATE CS | Propositional Logic| Weekly Quiz 4)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (63) out of 634 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Mathematical Logic

Let p and q be two propositions. Which of the following statements is a tautology /are tautologies?

A. $p \rightarrow p$
B. $(p \rightarrow p) \vee (p \rightarrow \bar{p})$
C. $(p \rightarrow p) \rightarrow (p \rightarrow \bar{p})$
D. $(p \vee \bar{q}) \vee \bar{p}$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Select Type | Award: 1 | Penalty: 0 | Mathematical Logic

Let $p = T, q = F, r = T$.
Select ALL expressions that evaluate to FALSE:

A. $q \rightarrow ((p \wedge \neg q) \vee r)$
B. $(\neg \mathrm{r} \vee \mathrm{q}) \leftrightarrow (\neg \mathrm{p} \wedge \mathrm{q})$
C. $\neg q \oplus r$
D. $\neg q \vee r$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Let the propositional variables p, q, and r represent the propositions:
p : You won the game
q : You scored the most points
r: You had the most fun
Select the logical expression that represents the statement: "**To have the most fun, it is necessary that you win the game**"

A. $p \leftrightarrow q$
B. $q \rightarrow p$
C. $r \rightarrow p$
D. $p \rightarrow q$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Find the assignment of truth values to p and q that prove that argument below is invalid.

$$\begin{array}{l} \mathbf{p} \vee \neg \mathbf{q} \\ \mathbf{q} \rightarrow \mathbf{p} \\ \hline \therefore \mathbf{p} \end{array}$$

A. $\mathrm{p} = \mathrm{T}, \mathrm{q} = \mathrm{T}$
B. $\mathrm{p} = \mathrm{T}, \mathrm{q} = \mathrm{F}$
C. $\mathrm{p} = \mathrm{F}, \mathrm{q} = \mathrm{T}$
D. $p = F, q = F$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 06sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

a. Given that the value of $p \rightarrow q$ is false, determine the value of $(\bar{p} \vee \bar{q}) \rightarrow q$.
b. Given that the value of $p \rightarrow q$ is true, can you determine the value of $\bar{p} \vee (p \leftrightarrow q)$?

Which of the following is the correct answer for (a), (b) respectively?

A. True, Yes
B. True, No
C. False, Yes
D. False, No

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

Let p, q and r be three propositions. Which of the following statements is a contingency /are contingencies?

A. $(p \vee \bar{q}) \rightarrow \bar{p}$
B. $p \leftrightarrow (\bar{p} \vee \bar{q})$
C. $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$
D. $(\bar{q} \rightarrow \bar{p}) \rightarrow (p \rightarrow q)$

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

You encounter a problem on an exam that is phrased as a multiple-choice question and you are told there is only one valid choice. Unfortunately, the question has been omitted! Here are the choices:

X. Claim A
Y. Claim A or claim B
Z. Claim B or claim C

Clearly, the questions included 3 claims that may, or may not have been true; however, as the question is missing, you do not know what those claims are!
Can you still find out what the correct choice is?

A. X
B. Y
C. Z
D. No, can't find out.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Consider the following statement:
If $\mathbf{x} < \mathbf{10}$ and $\mathbf{y} \geq \mathbf{0}$ then $\mathbf{x} < \mathbf{y} + 10$
If we wanted to prove this statement with a proof by contraposition, what assumption would we begin our proof with?

A. $x < 10$ and $y \geq 0$ and $x < y + 10$
B. $\mathrm{x} < 10$ and $\mathrm{y} \geq 0$
C. $x < y + 10$
D. $x \geq y + 10$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Mathematical Logic

From the expressions below, select ALL of the contradictions:

A. $r \leftrightarrow \neg r$
B. $(p \oplus q) \rightarrow (p \vee q)$
C. $(p \vee q) \rightarrow (p \oplus q)$
D. $(p \wedge q) \wedge (p \oplus q)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Mathematical Logic

Select the logical expression that is equivalent to:
$\neg(\mathbf{p} \wedge \mathbf{q}) \wedge (\neg\mathbf{p} \vee \mathbf{r})$

A. $p \wedge (\neg q \vee \neg r)$
B. $\neg(p \wedge q \wedge \neg r)$
C. $p \vee (q \wedge r)$
D. $\neg p \vee (\neg q \wedge r)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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